DO NOW — RECALL

# Timed Paper 1 data and logic practice

Name: _____ Class: _____ Date: _____

**1. Which sort builds an ordered prefix by inserting a key?**

**2. How do you compensate for shifting a positive fractional mantissa left?**

**3. Simplify ¬(A ∨ B).**

**4. Find the decimal value of $0.1100_2 \times 2^3$.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Insertion sort.

### 2. Model answer

Decrease the exponent by the same number of places.

### 3. Model answer

¬A ∧ ¬B.

### 4. Model answer

0.75×8=6.

### Vocabulary — know these terms

sign extension — Copy the sign bit into a new high-order position.

De Morgan — Swap AND/OR while negating components.

normalisation — Remove redundant leading mantissa sign bits.

mantissa — The fractional significant part.

exponent — Power-of-two scale.

NEW LEARNING

# Read, annotate and explain

### Show positive first

Use exactly eight bits for +11, rather than padding after converting.

### Invert each bit

00001011 becomes 11110100.

### Add one

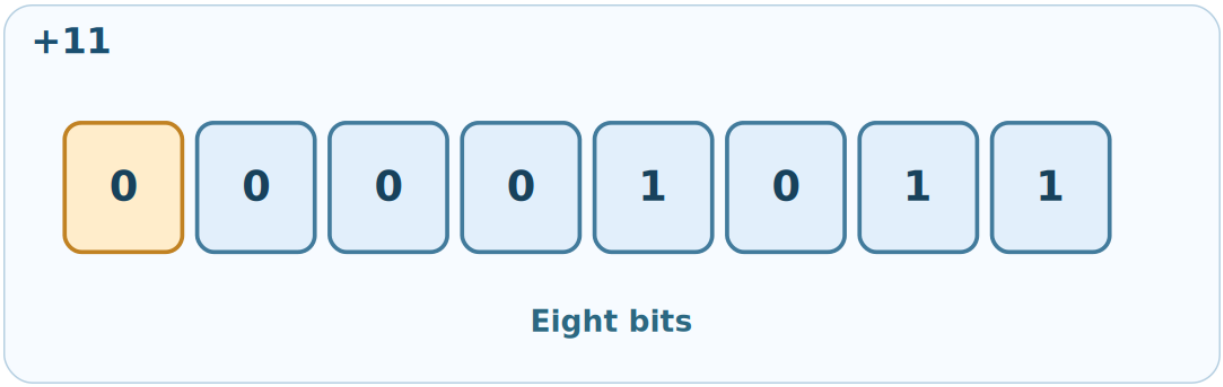
11110100 + 1 = 11110101 for −11.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

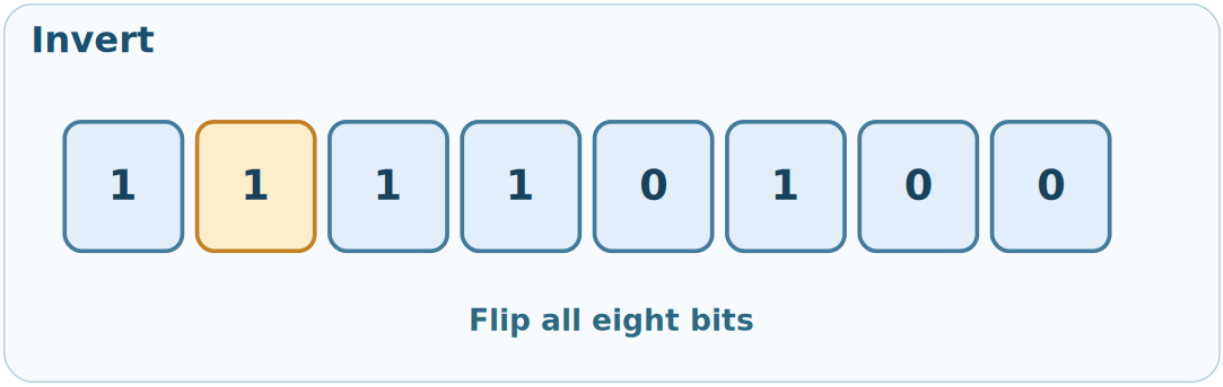
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




+11 0, 0, 0, 0, 1, 0, 1, 1 Eight bits

### New Learning

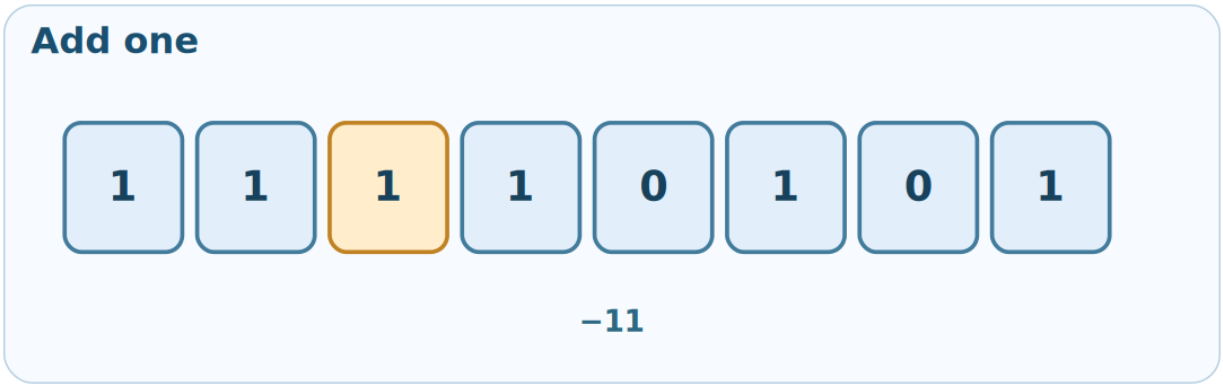



Invert 1, 1, 1, 1, 0, 1, 0, 0 Flip all eight bits

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Add one 1, 1, 1, 1, 0, 1, 0, 1 −11

I DO — WORKED EXAMPLE

# Follow the worked method

### Model −6 in eight-bit two's complement.

### Predict first

Write +6 as 00000110.

### Check the method

Invert to 11111001.

### Check the method

Add one to get 11111010, representing −6.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Model −6 in eight-bit two's complement.**

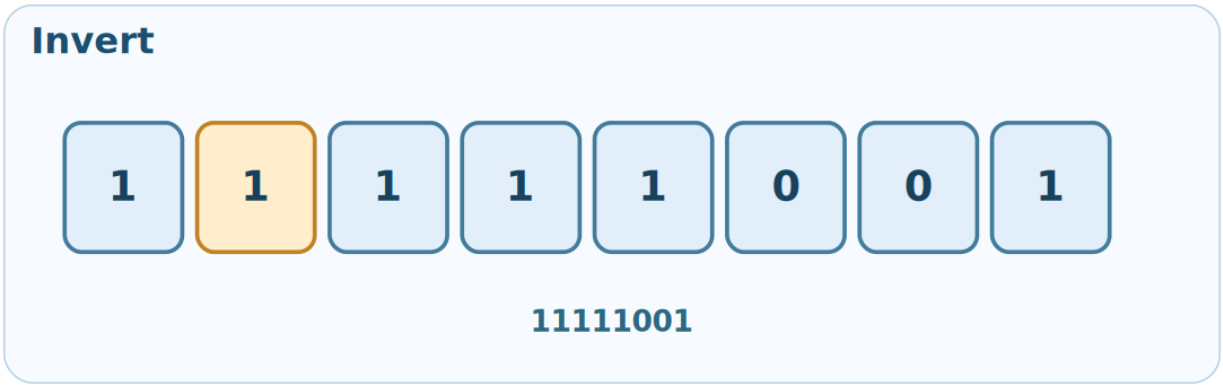
Write +6 as 00000110.




+6 0, 0, 0, 0, 0, 1, 1, 0 Positive

**Model −6 in eight-bit two's complement.**

Write +6 as 00000110.



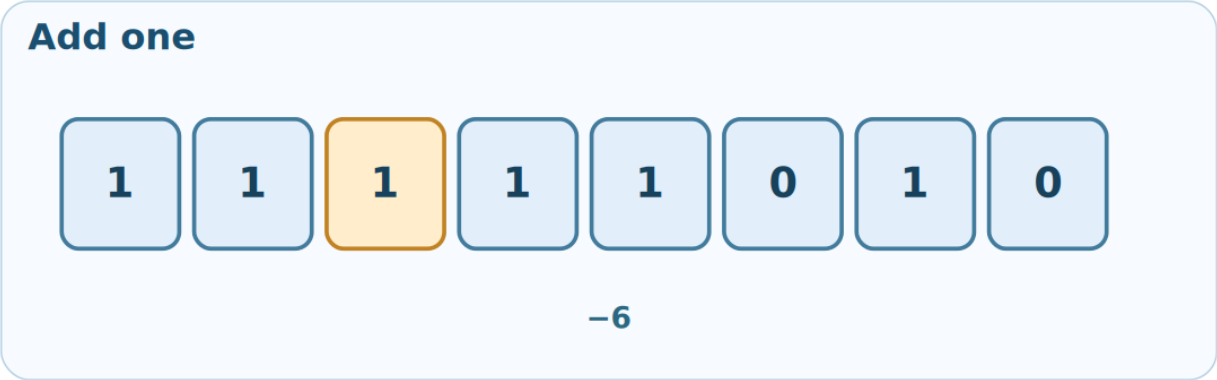

Invert 1, 1, 1, 1, 1, 0, 0, 1 11111001

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Model −6 in eight-bit two's complement.**

Write +6 as 00000110.




Add one 1, 1, 1, 1, 1, 0, 1, 0 −6

WE DO — GUIDED PRACTICE

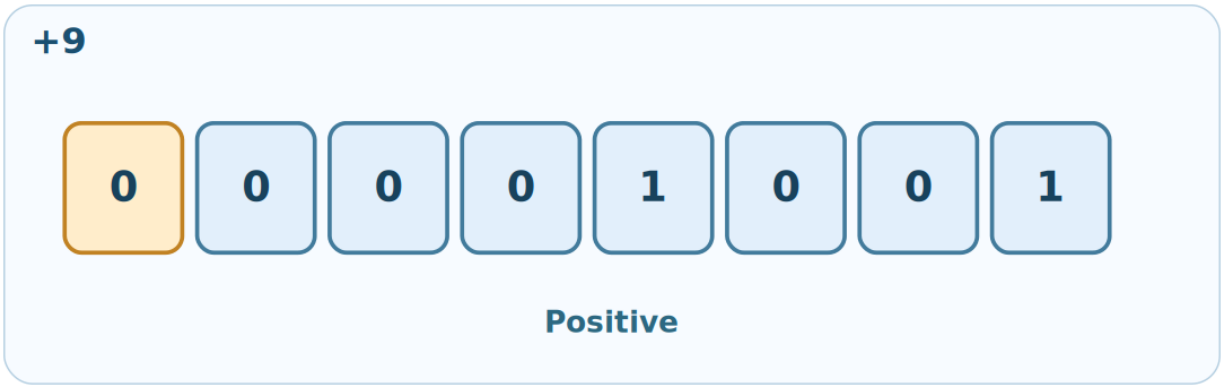
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Find −9 in eight-bit two's complement.**

Write 9 as 00001001.




+9 0, 0, 0, 0, 1, 0, 0, 1 Positive

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first Write 9 as 00001001.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

(a) Represent −19 as an 8-bit two's-complement number. [4] (b) Perform an arithmetic right shift by 1 of your result. State its signed decimal interpretation. [4]

### Question 2

(a) Use De Morgan's law to simplify ¬(A ∧ ¬B). [3] (b) Explain how an eight-cell group of ones simplifies a three-variable Karnaugh map. [5]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A positive two's-complement fractional mantissa is $0.0011_2$ with exponent 5. Normalise it, explaining each shift and why the value is unchanged.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An arithmetic right shift of −19 by one place produces −9.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

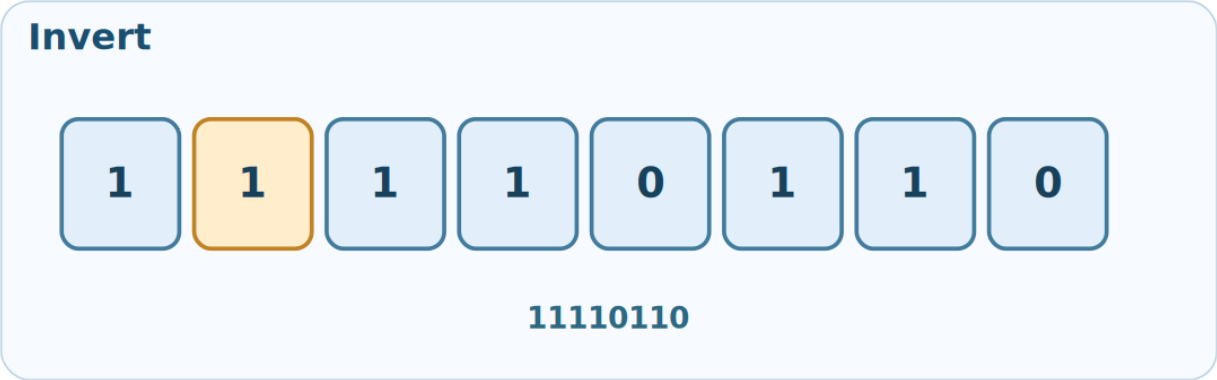
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Find −9 in eight-bit two's complement.**

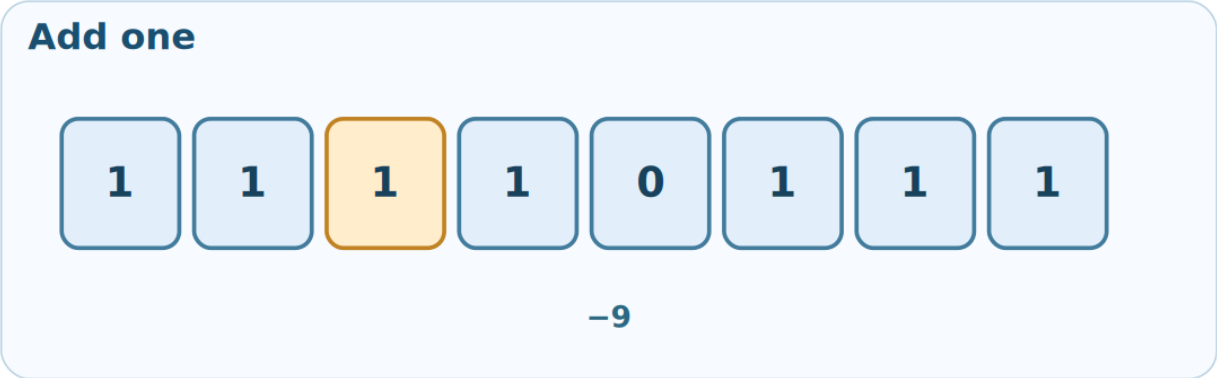
Write 9 as 00001001.




Invert 1, 1, 1, 1, 0, 1, 1, 0 11110110

**Find −9 in eight-bit two's complement.**

Write 9 as 00001001.




Add one 1, 1, 1, 1, 0, 1, 1, 1 −9

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Check the method

Invert to 11110110.

Add one to get 11110111, representing −9.

IMPROVE: Explain one correction to your own reasoning.